

Robotis (108490.KQ): Microduck takeoff highlights emerging physical AI demand; Robotis a key beneficiary

Pollen Robotics (part of Hugging Face) opened pre-orders for Microduck, a 25cm-tall open-source bipedal robot designed for physical AI and reinforcement learning applications. Priced at USD399, Microduck features 15 servo motors, onboard vision, and pre-trained locomotion behaviors, generating over 10,000 pre-orders within days of its August 27, 2026 launch. The company has been expanding its low-cost robotics lineup following the July 9, 2025 launch of Reachy Mini, an open-source desktop robot designed for human-robot interaction, priced at USD299. Unlike Reachy Mini, which was primarily designed for visual and audio interaction with users, Microduck focuses on locomotion, robot behavior learning and sim-to-real deployment, reflecting growing developer interest in low-cost robotics platforms. The strong initial demand shows accelerating developer interest in low-cost hardware for real world robot training, in our view.

Notably, each Microduck unit features 15 Robotis XL330 actuators, while Reachy Mini features 9 XL330 actuators, positioning Robotis as a direct beneficiary of both platforms' commercial success. Fulfillment of the initial Reachy Mini (20k units) and Microduck (10k units) production volumes implies incremental demand for c.330k actuators, equivalent to 80% of Robotis' 2025 annual order intake (410k units). On our estimates, this could generate c.W6.6bn of incremental revenue versus the 2025 revenue base of W38.9bn. Put differently, the incremental demand represents c.110% of Robotis' 2025 actuator capacity (300k units) and c.66% of its 2026E capacity (500k units), highlighting the scale of emerging physical AI demand.

Beyond immediate unit shipments, we believe Microduck's USD399 price point and open-source architecture lower barriers to real-world robotics experimentation, supporting broader developer participation in robotics training and experimentation. Microduck also supports a sim-to-real workflow, enabling policies trained in simulation to be deployed on physical robots, an increasingly important capability in physical AI development. This initiative aligns directly with Hugging Face's LeRobot ecosystem, where models, datasets and workflows are shared across an open robotics framework. As part of this effort, Hugging Face has also been integrating NVIDIA's GR00T humanoid foundation models into LeRobot, further expanding the tools available for physical AI development.

We believe the introduction of Microduck could broaden access to real-world robotics development through lower robot hardware acquisition costs. Since Microduck's sim-to-real software pipeline is built around XL330 performance characteristics, it creates a longer-term adoption opportunity for Robotis in addition

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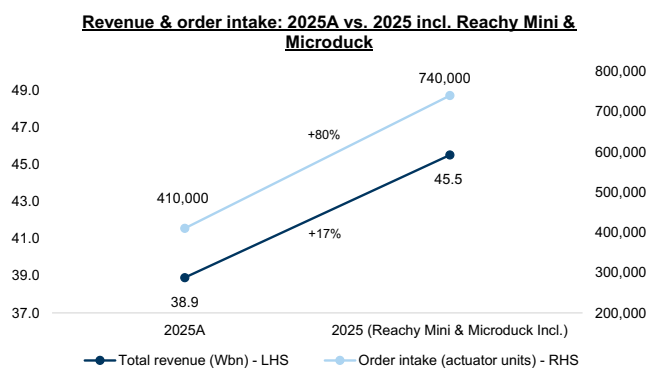
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to near-term hardware demand. Participation in Reachy Mini and Microduck also extends Robotis’ presence within the emerging open robotics ecosystem, while adding to its growing portfolio of global references including leading Chinese and US humanoid players.

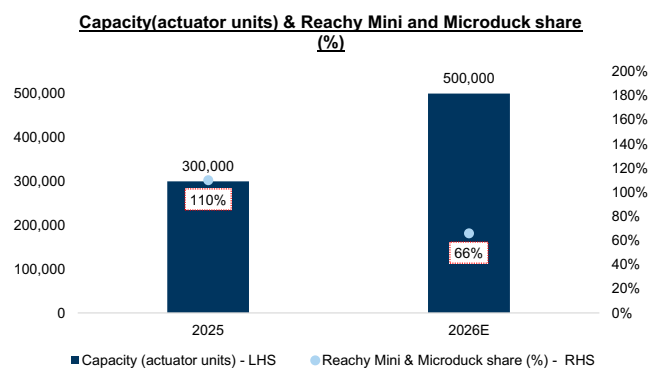
Interestingly, our estimates suggest that Robotis actuator content value comes at roughly USD132 per Reachy Mini unit (c.44% of retail price) and USD221 per Microduck unit (c.55% of retail price), suggesting actuators remain a major cost component for humanoid robots. Hence we continue to believe Robotis is well positioned to benefit as humanoid adoption scales and the physical AI market grows over time. We are Buy-rated on Robotis.

Exhibit 1: Initial Reachy Mini and Microduck demand alone could add c.W6.6bn of revenue and expand order intake by 17%...



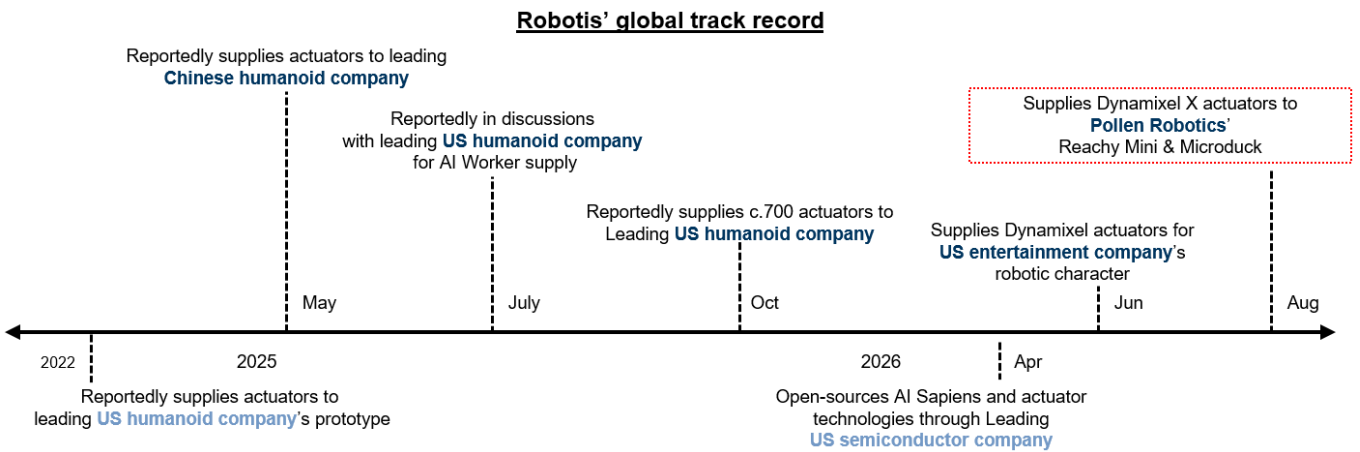
Source: Various media reports, Company data, Goldman Sachs Global Investment Research

Exhibit 2: ...while consuming ~66% of current actuator capacity



Source: Various media reports, Company data, Goldman Sachs Global Investment Research

Exhibit 3: Expanding global customer footprint across leading humanoid and physical AI platforms



Includes key track records reported in media and hence may not be exhaustive

Source: Various media reports, Goldman Sachs Global Investment Research

Price Target Risks and Methodology - Robotis

We are Buy rated. Our 12-month target price of W630,000 is based on a target P/E multiple of 60.0x applied to 2035E EPS discounted back to June 2027.

We see three downside risks: 1) Execution risks due to slower progress at the Uzbekistan

facility, 2) No major orders for humanoid actuators in 2027/2028, and 3) Aggressive pricing on key products leading to dilution in margins and earnings.

108490.KQ	12m Price Target: W630,000	Price: W248,500	Upside: 153.5%
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Buy		GS Forecast			
Market cap: W3.6tr / \$2.7bn Enterprise value: W3.5tr / \$2.5bn 3m ADTV: W57.0bn / \$38.8mn South Korea Korea Autos & Industrials M&A Rank: 3 Leases incl. in net debt & EV?: Yes	Revenue (W bn)	12/25	12/26E	12/27E	12/28E
	EBITDA (W bn)	38.9	98.3	200.5	307.2
	EPS (W)	5.9	8.6	40.1	65.2
	P/E (X)	346	306	1,712	2,731
	P/B (X)	NM	NM	145.1	91.0
	Dividend yield (%)	4.7	11.4	10.6	9.5
	N debt/EBITDA (ex lease,X)	0.0	0.0	0.0	0.0
	CROCI (%)	(39.0)	(20.1)	(3.5)	(1.8)
	FCF yield (%)	4.1	3.3	13.4	21.3
		0.3	(0.7)	(0.0)	0.2
		3/26	6/26E	9/26E	12/26E
	EPS (W)	(691)	172	229	596

Source: Company data, Goldman Sachs Research estimates, FactSet. Price as of 3 Sep 2026 close.

Disclosure Appendix

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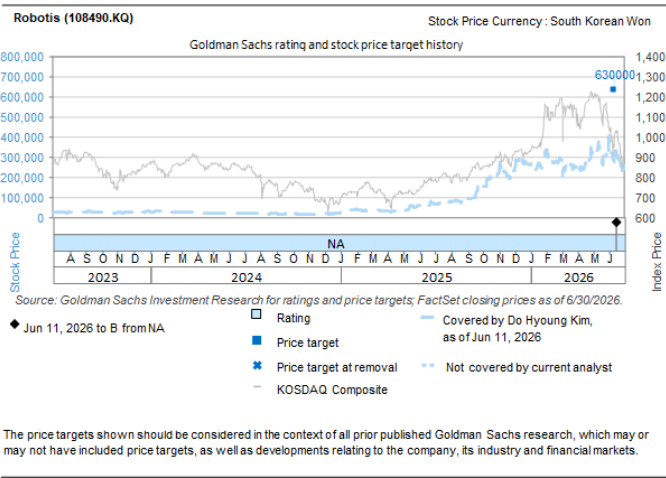
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Price target and rating history chart(s)



Target price history table(s)

Robotis (108490.KQ)

Date of report	Target price (W)	Closing price (W)
11-Jun-26	630,000	286,500

Price targets shown in table(s) are unadjusted for corporate actions.

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